

Practical Task (4 Hours)

Build a Scalable Product Management API with DRF, PostgreSQL, Celery, and Redis

- **Project Overview**

- Develop a RESTful API using Django Rest Framework (DRF) with the following features:
 1. Custom User Authentication (Custom User Model, Login, Logout, Password Reset)
 2. Category & Product CRUD (Admin Only)
 3. Soft Delete Instead of Permanent Deletion
 4. Bulk Upload of Products & Categories from JSON File
 5. Processed upload in Background using Celery (*Optional - Plus Point*)
 6. Provide API collection
 7. Provide requirements file

- **Technology Stack**

- Backend: Django, Django Rest Framework (DRF)
- Database: PostgreSQL
- Git - maintain commits after module/task done

Features & Implementation Details

1. Authentication & User Management

- Custom User Model (Email-based authentication instead of username - *Optional - Plus Point*)
- Endpoints:
 - POST /auth/register/ → Register a user
 - POST /auth/login/ → Login
 - POST /auth/logout/ → Logout (Blacklist token)
 - PATCH /auth/update/ → Update user details
 - POST /auth/forgot-password/ → Send password reset link via email (*Optional - Plus Point*)

2. Category & Subcategory Management

- Category Model
- Admin CRUD Only
- Endpoints:
 - POST /category/ → Create category (Admin Only)
 - GET /category/ → List all categories (Public)
 - GET /category/{id}/ → Retrieve a specific category
 - PATCH /category/{id}/ → Update category (Admin Only)
 - DELETE /category/{id}/ → Soft delete category (Admin Only)

3. Product Management

- Product Model (Belongs to a Category)
- CRUD only for Admin, Read for All Users
- Endpoints:
- POST /product/ → Create a product (Admin Only)
- GET /product/ → List all products (Public)
- GET /product/{id}/ → Retrieve a specific product
- PATCH /product/{id}/ → Update product details (Admin Only)
- DELETE /product/{id}/ → Soft delete product (Admin Only)

4. Bulk Upload Categories & Products from JSON File - (Must)

- Users can upload a JSON file containing category and product data
- Processing is handled asynchronously using Celery to avoid request timeouts

(Optional - Plus Point)

- File structure example:

```
categories=[
  {
    "id": 1,
    "category_name": "Electronics",
    "description": "Devices including smartphones, laptops, and accessories",
    "created_at": "2025-02-20T12:00:00Z",
    "updated_at": "2025-02-20T12:00:00Z"
  },
  {
    "id": 2,
    "category_name": "Clothing",
    "description": "Fashion items including shirts, pants, and accessories",
    "created_at": "2025-02-20T12:00:00Z",
    "updated_at": "2025-02-20T12:00:00Z"
  }
]

products=[
  {
    "id": 101,
    "category_id": 1,
    "category_name": "Electronics",
```

```

        "product_name": "Smartphone",
        "product_description": "Latest model with high-resolution camera and OLED
display",
        "product_price": 699.99,
        "currency": "INR",
        "stock_quantity": 50,
        "sku": "ELEC-SMART-001",
        "image_url": "https://example.com/smartphone.jpg",
        "created_at": "2025-02-20T12:00:00Z",
        "updated_at": "2025-02-20T12:00:00Z"
    },
    {
        "id": 102,
        "category_id": 2,
        "category_name": "Clothing",
        "product_name": "T-Shirt",
        "product_description": "100% cotton, comfortable fit",
        "product_price": 19.99,
        "currency": "INR",
        "stock_quantity": 200,
        "sku": "CLOTH-TSHIRT-001",
        "image_url": "https://example.com/tshirt.jpg",
        "created_at": "2025-02-20T12:00:00Z",
        "updated_at": "2025-02-20T12:00:00Z"
    }
]

```

- Endpoints:
- POST /upload/ → Upload JSON file (Admin Only)
- Celery task picks up the file, processes it, and adds data to the database(*Optional - Plus Point*)

5. Soft Delete Instead of Permanent Deletion

- Instead of deleting records, use a `is_deleted=True` flag
- Filter out soft-deleted records in API responses

6. Git

Testing Strategy

Unit Tests

- Test authentication (User registration, login, logout)
- Test category creation, retrieval, update, soft delete
- Test product creation, retrieval, update, soft delete

Integration Tests

- Ensure user authentication and permissions work correctly
- Check if category-product relationships are maintained
- Verify that the JSON upload process correctly adds data to the database